MaleVis Task 1 Report

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1. Introduction

Malware classification is a central cybersecurity problem because malicious files continuously evolve through repacking, obfuscation, and code modification. Signature-based detection models can be ineffective when new variants do not exactly match known signatures. Image-based malware analysis provides an alternative representation in which software byte sequences are mapped directly to multi-channel pixel profiles. This visual transformation allows convolutional neural networks to learn recurring texture and structural patterns without requiring slow, complex disassembly or code execution. Unlike early grayscale approaches, advanced methods deploy three-channel RGB color mapping to preserve deep, multi-dimensional byte relationships.

This project follows the Custom CNN + Attention track to classify software categories using the MaleVis dataset. Task 1 explores the dataset structure, analyzes multi-class distributions across malware families and benign software classes, and evaluates pixel-level texture quality metrics. This analysis establishes a clear research gap centered on handling the redundant byte-image problem caused by static padding artifacts. The later tasks will develop representative custom CNN baselines, implement a lightweight spatial/channel attention module to filter out uninformative canvas padding, execute controlled model ablation experiments, and conduct behavioral interpretability analyses.

2. Dataset Description

The MaleVis Dataset represents software binaries as multi-channel RGB color images to capture advanced textural layouts. The canonical corpus utilized in this project comprises a total of 14,226 images categorized across 26 distinct software classes. This distribution includes 25 malicious malware families alongside 1 legitimate benign software class labeled as Other. Every individual image sample is stored within a class-specific folder structure, allowing directory paths to be systematically targeted for multiclass label generation.

Table 1: Core properties of the MaleVis dataset.

Property	Description
Dataset name	Malevis
Source	Kaggle
Domain	Cybersecurity
Learning task	Multiclass malware and benign software classification
Standard image count	14,226
Number of classes	26 (25 Malware families + 1 Benign class)
Image representation	Three-channel RGB color byte images
Dominant file format	PNG
Repository dataset policy	Dataset files are not committed to GitHub

3. Exploratory Data Analysis Methodology

The EDA notebook was designed to satisfy the required image-analysis items while preserving reproducibility and preventing avoidable memory usage. The main steps are summarized below.

- Recursively enumerate supported image files and infer labels from parent folders.
- Validate every image using PIL and OpenCV and record unreadable files separately.
- Extract width, height, aspect ratio, resolution, file format, image mode, channel count, and file size.
- Measure multi-channel color brightness, contrast, Laplacian variance, and Shannon entropy.
- Calculate class counts, percentages, the majority-to-minority ratio, and cumulative class concentration.
- Display one labelled sample from every malware family and benign software class, alongside multiple samples from selected majority and minority classes.
- Sample pixel intensities efficiently to avoid loading every image pixel into memory.
- Detect exact duplicate files using SHA-256 hashing and identify any cross-class duplicate groups to prevent data leakage before model splitting.
- Export report-ready figures and CSV tables to the designated local workspace output directory.

4. Class Distribution and Imbalance

The dataset is highly imbalanced. The largest class, Other (representing benign software samples), contains 1,832 images, whereas the smallest malware family, Agent, contains only 470 images. The resulting majority-to-minority ratio is 3.90:1. The Other class accounts for a substantial

proportion of the dataset, which means that overall accuracy can be easily dominated by this single benign class.

Table 2: Dataset imbalance metrics for the MaleVis corpus.

Measure	**Result**
Total images	14,226
Total classes	26 (25 Malware + 1 Benign)
Majority class	Other (Benign) (1,832)
Minority class	Agent (480)
Majority-to-minority ratio	3.90:1
Primary evaluation emphasis	Macro-F1 and class-wise recall

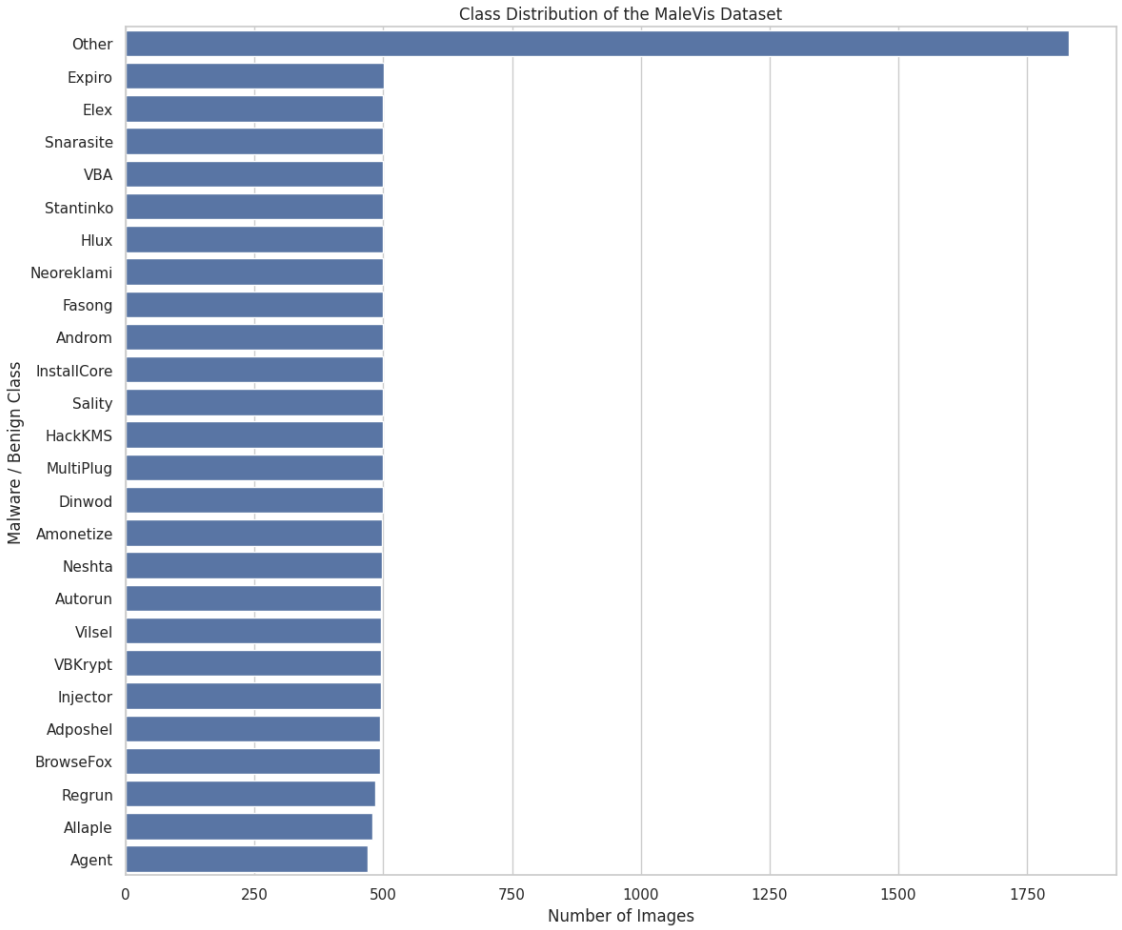


Figure 1: Class distribution of the standard MaleVis corpus.

Because of this imbalance, Task 2 should not rely on accuracy alone. The evaluation must include macro precision, macro recall, macro-F1, weighted F1, class-wise metrics, and a confusion matrix.

Class weights or another carefully justified imbalance-handling method should be applied only to the training data.

5. Labelled Image Analysis

A labelled grid with one representative image from every class is used to compare broad visual patterns. Malware byte images often contain repeated horizontal bands, dense texture blocks, sparse regions, and family-specific structures. Some families appear visually distinct, while others share similar multi-channel layouts or noisy grayscale profiles. These visually similar classes are likely to produce confusion and are important targets for later class-wise error analysis.

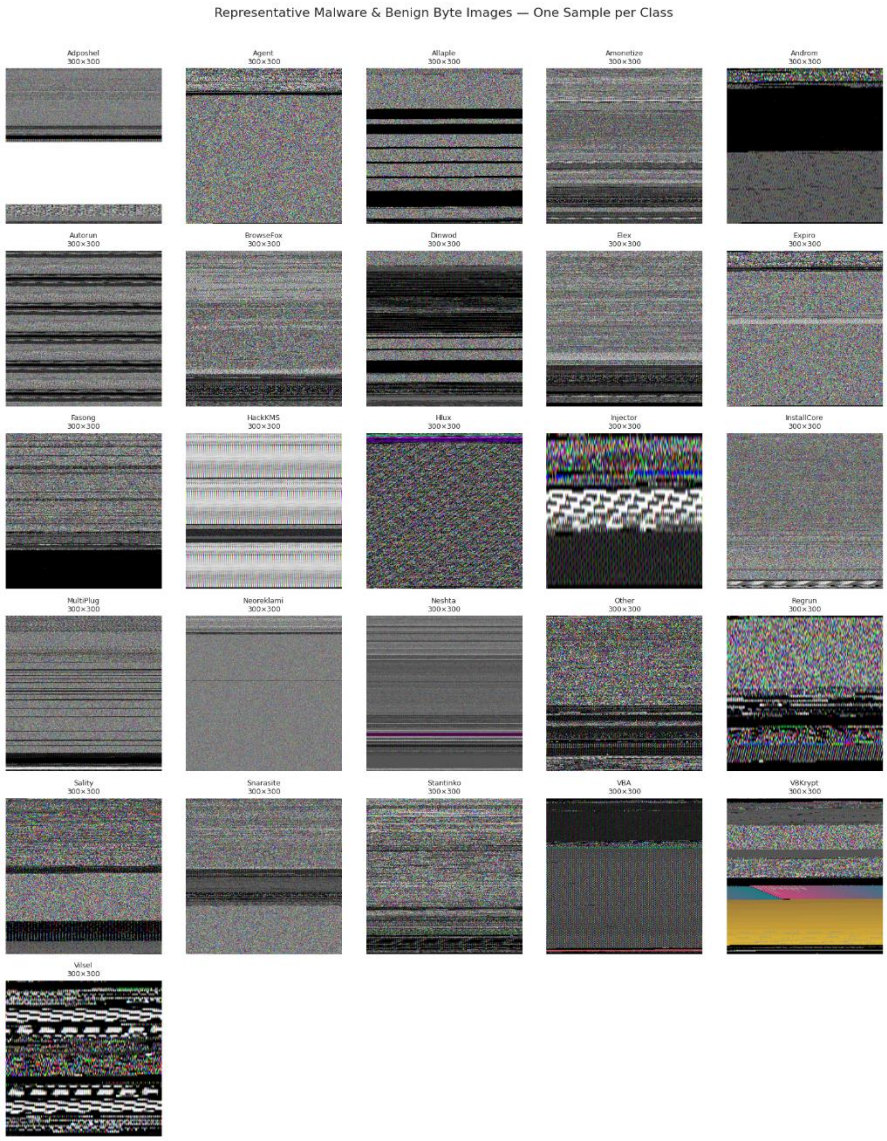


Figure 2: One representative software byte image from each class.

A second grid is used to compare several examples from selected large and small classes. This comparison reveals whether a class has a consistent internal texture or exhibits strong within-class variation. The observed variation supports the use of an attention mechanism, as the proposed network must learn to emphasize discriminative feature channels and critical spatial regions rather than treating every pixel location equally.

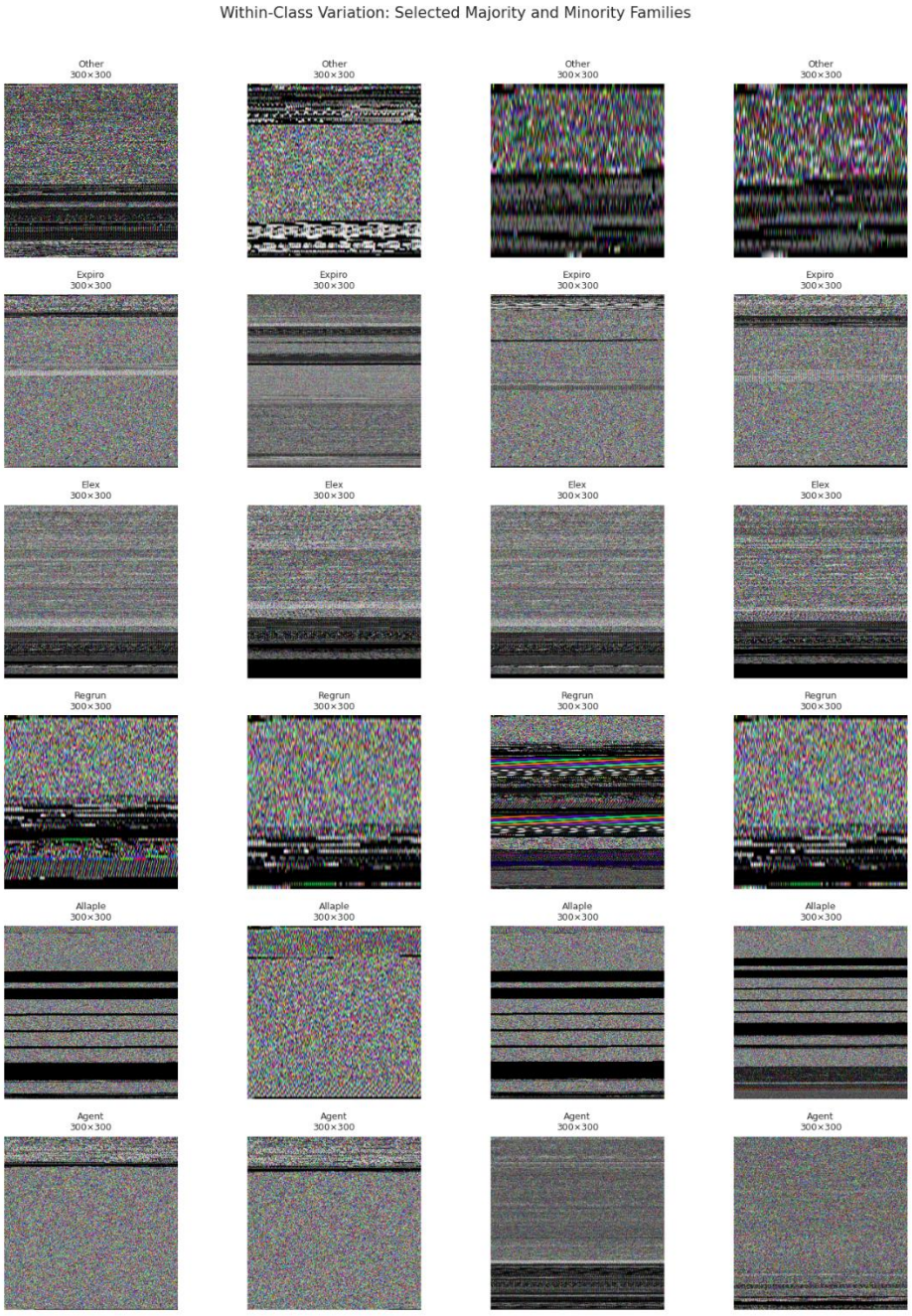


Figure 3: Within-class visual variation for selected majority and minority families.

END